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**Analyzing Issue Discussions in Open Source Software Communities: Unveiling
the Characteristics of Usability Issues**

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Thèse présentée en vue de l'obtention du diplôme de *Philosophiæ Doctor*
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Cette thèse intitulée :

**Analyzing Issue Discussions in Open Source Software Communities: Unveiling
the Characteristics of Usability Issues**

présentée par **Arghavan SANEI**

en vue de l'obtention du diplôme de *Philosophiæ Doctor*
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DEDICATION

This thesis is dedicated to my beloved husband Nima Kiakojoori, whose unwavering support and encouragement have been my strength throughout this journey. To my dear parents Shohreh and Masoud, whose love, sacrifices, and guidance have shaped me into who I am today. And to my brother Aria, whose constant belief in me has been a source of inspiration.

Thank you all for your endless love and support.

PREVIEW

ACKNOWLEDGEMENTS

The journey of pursuing my PhD was undoubtedly challenging. It tested me intellectually, mentally, and emotionally. The path was filled with obstacles, confusion, self-doubt, self-criticism, failures, tears, and breakdowns. Many times, I questioned whether to continue this journey and whether to change my extremely challenging research topic. The first two years were particularly tough, as I struggled to research the impact of collaboration in software discussions with no success, especially during the isolating times of COVID-19. It was very demotivating! However, looking back and connecting the dots, I am glad I persevered and did not give up.

A turning point came when a contributor on emotional states was accepted into the Cooperative and Human Aspects of Software Engineering (CHASE) conference. This milestone validated my focus on human impacts in software discussions and renewed my hope. This PhD thesis is my modest contribution to understanding how humans influence discussions with their emotions, opinions, intentions, and arguments. I never imagined I would finish my PhD in less than five years, complete two internships in Natural Language Processing, and have my thesis accepted as it is. Of course, a PhD is not a solo endeavor. This achievement is partly due to the wonderful people who supported me along the way.

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RÉSUMÉ

Le développement des logiciels Open Source (OSS) repose largement sur des discussions collaboratives entre des participants divers, facilitées par des systèmes de suivi des problèmes (ITS) tels que GitHub. Ces types de plateformes révolutionnent la gestion des bugs, soutiennent les discussions entre participants divers et convertissent les idées et les problèmes en tâches concrètes. De plus, ces plateformes offrent des opportunités à des contributeurs divers, quels que soient leurs antécédents et leur expertise, pour s'engager et collaborer dans le développement de logiciels et la résolution de problèmes dans les projets OSS. Comprendre cette diversité en tant que comportement humain et interaction au sein des communautés OSS est utile pour une collaboration et une communication efficaces entre les participants, impactant la direction des projets, la centrabilité sur l'utilisateur et la qualité globale du projet. Cependant, les dynamiques de l'interaction humaine au sein des communautés OSS, en particulier dans les discussions sur les problèmes de convivialité, sont restées peu étudiées. De plus, l'accent prédominant sur les contributions techniques conduit souvent à négliger les aspects de convivialité, impactant la conception et la satisfaction des utilisateurs. Pour aborder ces problèmes, cette thèse, à travers des études empiriques, examine l'influence des problèmes et commentaires postés émotionnellement, les caractéristiques des problèmes de convivialité postés, la qualité et le discours des arguments, ainsi que les antécédents de l'expertise en UX sur les discussions sur la convivialité.

La première étude explore l'impact des sentiments et des tons incarnés dans les discussions générées par la communauté au sein des projets OSS. Selon l'analyse empirique, il y avait des corrélations entre les sentiments, les tons et diverses mesures de discussion, offrant des implications pour les contributeurs OSS et les concepteurs d'outils. La prochaine étude de cette thèse caractérise les discussions sur les problèmes de convivialité dans les projets OSS, en se concentrant sur les sujets prédominants, les dimensions de convivialité et les dynamiques collaboratives dans les discussions sur la convivialité. Les résultats ont indiqué des caractéristiques distinctives des discussions sur la convivialité et ont souligné l'importance de traiter les problèmes de convivialité pour améliorer l'efficacité globale du projet. Ensuite, la thèse analyse le discours et la qualité des arguments dans les discussions sur les problèmes de convivialité, révélant le rôle significatif de l'argumentation dans la formation des caractéristiques de la discussion. Les insights de cette étude fournissent des recommandations pratiques pour soutenir des arguments efficaces et améliorer les pratiques de convivialité OSS. De plus, dans l'étude finale, elle examine les caractéristiques des problèmes de convivialité postés par les contributeurs, en particulier les professionnels de l'UX, en soulignant leur approche distinc-

tive pour rapporter les problèmes de convivialité et la rareté de leur collaboration dans les projets OSS. Cela souligne le besoin d'une plus grande collaboration et d'une invitation pour les professionnels de l'UX afin d'améliorer la convivialité et la qualité de conception dans les projets OSS.

Dans l'ensemble, cette thèse contribue à une compréhension plus profonde et plus complète de la nature dynamique des discussions de développement OSS et de l'importance de traiter les problèmes de convivialité pour améliorer les expériences utilisateur et la qualité des projets. En fournissant des insights, les projets OSS peuvent élaborer des recommandations pratiques.

PREVIEW

ABSTRACT

The development of Open Source Software (OSS) highly relies on collaborative discussions among diverse participants, facilitated by Issue Tracking Systems (ITSs) such as GitHub. These types of platforms revolutionize handling bugs, supporting diverse participants' discussions and converting ideas and issues into actionable tasks. Also, these platforms offer opportunities to diverse contributors with any backgrounds and expertise to engage and collaborate in software development and problem-solving for OSS projects. Understanding this diversity as human behaviour and interaction within OSS communities is helpful for effective collaboration and communication between participants, impacting project direction, user-centricity, and overall project quality

However, the dynamics of human interaction within OSS communities, particularly in usability issue discussions, stayed understudied. Also, the predominant focus on technical contributions often leads to neglecting usability aspects, impacting the design and user satisfaction. To address these, this thesis through empirical studies investigates the influence of emotionally posted issues and comments, characteristics of usability posted issues, argument quality as well as discourse, and the backgrounds of UX expertise on usability discussions.

The first study explores the impact of sentiments and tones embodied in community-generated discussions within OSS projects. According to the empirical analysis, there were correlations between sentiments, tones and various discussion metrics, offering implications for OSS contributors and tool designers. The next study of this thesis characterizes usability issue discussions in OSS projects, focusing on prevalent topics, usability dimensions and collaborative dynamics in usability discussions. The results indicated distinctive characteristics of usability discussions and highlighted the importance of addressing usability concerns to enhance overall project effectiveness. Subsequently, the thesis analyzes argument discourse and quality in usability issue discussions, revealing the significant role of argumentation in shaping discussion features. Insights from this study provide practical recommendations to support effective arguments and enhance OSS usability practices. Furthermore, in the final study, it examines the characteristics of posted usability issues by contributors, in particular, UX professionals, emphasizing their distinctive approach to reporting usability issues and the scarcity of their collaboration in OSS projects. This highlights the need for greater collaboration and invitation on UX professionals to improve usability and design quality in OSS projects.

Overall, this thesis contributes to a deeper understanding and comprehension of the dynamic

nature of OSS development discussions and the importance of addressing usability concerns to enhance user experiences and project quality. By providing insights, OSS projects can develop practical recommendations.

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PREVIEW

LIST OF SYMBOLS AND ACRONYMS

OSS	Open Source Software
ITS	Issue Tracking systems
UI	User Interface
UX	User eXperience

PREVIEW

CHAPTER 1 INTRODUCTION

1.1 Context and Problem Statement

Open Source Software (OSS) development offers numerous opportunities for public discussions, occurring across various platforms. These opportunities reflect the spirit of open-source development by allowing individuals with relevant expertise to influence the project's direction while enabling those without prior experience to collaborate and contribute to this process. To this point, within the ecosystem of OSS, Issue Tracking Systems (ITSs) like Jira and GitHub Issues have revolutionized the handling, resolving, and elaboration of community-generated topics, fostering comprehensive discussions among community members, and transforming emerging ideas within issues into actionable tasks for development teams [2, 3]. As contributors engage in these public discussions and conversations, they not only impact project features but also influence one another, emphasizing the dynamic nature of OSS development [4, 5].

The picture presented from previous research indicates that OSS not only thrives on the projects that this development model supports but also on the successful and growing communities. Effective communication and collaboration in these communities are fundamental to the success of OSS projects. The collaborators of these communities are a diverse range of stakeholders, including developers, maintainers, designers, and end-users whose interactions significantly impact the project's quality, direction, and lifecycle [6–8]. While the technical merit of the codebase is crucial, OSS development extends beyond only coding; addressing real-world problems, managing projects effectively, and prioritizing users' requirements are equally important [8, 9]. To this point understanding contributors' behaviour within issue discussion threads offers valuable insights into these dynamics. Each comment and response shape the project's path, influences the user experience, and highlights crucial concerns and expectations [4, 5, 7]. This, in turn, guides the development of user-centric software [10, 11]. Furthermore, the nature of these interactions directly affects project adaption and sustainability [12]. Positive, inclusive, persuasive communication fosters a welcoming community environment, attracting and retaining valuable contributors [13, 14]. In contrast, negativity and a toxic environment can discourage potential collaborators and disrupt community relations [15]. Therefore, extensive empirical studies on human interaction in a positive and persuasive manner in issue discussions are crucial. By identifying the factors that promote collaboration, inclusively, and positive communication, we can cultivate thriving OSS communities. This thesis can significantly impact the long-term success of the OSS ecosystem,

promote user-centric development practices, and enhance the overall discussion dynamics within these communities.

In this context, the diverse backgrounds and roles of community members who contribute to issue reporting and discussion play a significant role [4, 16, 17]. While this diversity in contributors of ITSs provides valuable perspectives for project advancement, consolidating and addressing these varied viewpoints within a large volume of community-generated requests and discussions presents a challenge. For example, gender, tenure, and technical background bring a wide range of insights and expertise to the projects, resulting in a more comprehensive understanding of end-users variety of requirements and presenting unique challenges for addressing different types of issues such as the usability of the projects. Moreover, this diversity in OSS community members results in a variety of valued factors, practices, and challenges concerning usability, leading to a more holistic approach to addressing usability issues in OSS projects [18]. Overall, the diverse backgrounds and contributions of OSS contributors bring valuable insights, varied approaches, and unique challenges to the forefront when addressing usability in OSS projects [4, 17, 19, 20]. Another example of the contributors' diversity is the impact on contributors from various geographical locations and diverse commit patterns which influence software quality improvements [21–23]. Also, gender diversity and employment diversity were found to be correlated with a reduction in software quality as diversity increased [22, 23]. This suggests that while certain types of diversity can enhance software quality, others may have a negative impact [22, 23]. Thus, understanding and facilitating effective discussion within ITSs is crucial for addressing the diverse needs and perspectives of OSS contributors.

At the same time, while OSS platforms welcome contributors from individuals of all backgrounds, the predominant focus on technical contributions often overshadows concerns related to UX design and the usability of the software [18, 24]. This skewed focus is often prevalent among many OSS contributors who come from technical backgrounds and prioritize functionalities and features over user experience and usability [18]. Consequently, usability emerges as a critical but overlooked aspect related to the OSS projects, limiting their ability to provide optimal user experiences [25, 26]. This neglect results in insufficient usability, which in turn affects the overall effectiveness of OSS applications and user satisfaction [26, 27]. Platforms like ITSs serve as collaborative spaces where community members can report and discuss various topics related to OSS development, including usability issues [18, 28]. By facilitating direct connections between users, designers, developers, and maintainers, ITSs play a pivotal role in fostering comprehensive discussions and supporting mutual contributions to OSS projects. However, the challenge remains in effectively integrating diverse perspectives and expertise to address usability issues and elevate the overall user experience within OSS

communities.

Previous studies have established that discussion participants are influenced by the **emotion and affective states** expressed in the discussion [29, 30]. In the context of software development, emotions and affective states can be defined as the psychological experiences that influence developers' behavior, motivation, and performance. These factors can be expressed through different types of tones and measured through the dimensions of Valence, Arousal, and Dominance (VAD), which quantify the attractiveness, activation level, and perceived control over a situation, respectively [30]. These emotional aspects provide valuable information that can be utilized by software development teams, such as predicting the time needed to complete a code review [31]. However, recent research has primarily focused on advancing automatic emotion detection techniques rather than exploring potential use cases for these techniques [32]. Studies on the effects of emotions and affective states in issue discussions have predominantly concentrated on development aspects, such as bug resolution time and commit size. The social and community aspects, such as the dynamics of follow-up discussions and responses, on the other hand, remain less explored. Understanding these social and community aspects is crucial because they significantly influence the overall quality and efficiency of the issue review and resolution process. Positive social interactions can persuade more collaboration in the community, leading to more effective communication and quicker resolution of issues [13, 14]. In contrast, negative interactions or toxic ones can hinder progress and create misunderstanding [15]. Additionally, emotions and affective states expressed in issue discussions are often connected with other factors, such as the nature and complexity of the issue which can impact both social and developmental aspects of OSS projects [33, 34]. By examining these social dimensions, we can gain insights into how emotional dynamics shape collaborative efforts, ultimately enhancing the effectiveness and sustainability of OSS projects. Therefore, a comprehensive understanding is still needed to assess the importance of these factors in how members of the OSS community react to and address issues.

The multifaceted collaboration within OSS communities, especially in usability issues, highlights the critical role of social and community aspects, such as follow-up discussions and responses. These aspects are crucial since they directly impact the collaborative dynamics and overall success of OSS projects. Effective communication had positive interactions foster a supportive environment, which enhances the collective problem-solving abilities of the community. In contrast, poor interactions can lead to misunderstanding and conflicts, hindering progress. Thus, diverse contributors use **persuasive argumentation** to influence and convince each other, improving usability and user experiences [35–37]. Understanding how persuasive argumentation works within these communities is vital because it shapes the decision-making process and the quality of the outcomes. Persuasive arguments help in

presenting different perspectives, negotiating solutions, and reaching consensus which are essential for addressing complex usability issues [38,39]. While previous research has examined the argumentation structure of usability issues [24], there is still much to be investigated regarding how the quality of arguments and collaborative behaviour of participants influence the discussions. These factors play a significant role in the effectiveness of communication and the ability to achieve desirable outcomes in OSS projects [38,39]. By delving into these community aspects and influences, we can better understand and enhance the collaborative processes that drive the success of OSS projects.

Apart from argument impacts in OSS discussion, **usability issue posters** also play a crucial role in enhancing the usability of open-source software by providing feedback, insights, and problem reports that contribute to user-centred improvements. Recognizing the unique characteristics and contributions of usability issue posters is essential for open-source communities to prioritize and address usability issues effectively. Existing research has primarily focused on the content and resolution of issues in these platforms, but less attention has been given to identifying and understanding those who are dedicated to addressing usability concerns [18,40–42].

To fill the research gaps mentioned above, this thesis explores how diverse contributors in OSS communities influence each other through emotional engagement, persuasive argumentation, and the utilization of their UX expertise to improve usability and user experiences.

Through an exploration of the discussion dynamics of OSS projects, this thesis aims to examine the impact of diverse backgrounds of collaborators and their varied viewpoints as emotions and arguments in posting issues. Furthermore, this thesis delves into the critical yet overlooked aspect of usability in OSS projects, highlighting recent shifts in focus toward improving usability within collaborative platforms like GitHub. These investigations contribute to a deeper understanding of the dynamic nature of OSS development and the importance of addressing usability concerns to enhance user experiences and overall project effectiveness.

1.2 Thesis Overview and Organization

In the following, the overview of the thesis is presented:

Chapter 2: *Background and related research*

We first offer the reader background information and introduce previous research on topics

related to this thesis, with respect to which this thesis is positioned. Then, we transition to the core body of this thesis, concentrating on reporting empirical studies. Each chapter delves into reporting methodology, results, discussion, and potential threats to the validity of the respective study.

Chapter 3: *The Impacts of Sentiments and Tones in Community-Generated Issue Discussions*

To aim toward investigating human behaviour and discussion dynamics in OSS, the first study explored the impact of sentiments and tones expressed in community-generated discussions within OSS projects. The complex affective states that diverse community members have exhibited in posting issues and participating in discussions significantly affect the efficacy of issue discussions in resolving and improving the software into actionable tasks for development purposes. Through an extended empirical study, we examined the correlations between sentiments, tones, and various discussion and development-related metrics across three prominent OSS projects. This research investigated how sentiments and tones in issue discussions impact factors like discussion length, closure time, and the roles as well as types of issues. We created and validated a substantial dataset comprising sentiments and tones from 32,764 issues and 192,956 comments, enabling a comprehensive analysis. Considering the multifaceted nature of these discussions, we controlled factors such as discussion length, commit size, issue reporter role, and issue type in the analysis. The findings of this study painted a picture of how affective states not only interact with these factors but also the relationships between them. Ultimately, this study's insights offer practical implications for OSS contributors and tool designers, informing effective issue discussion moderation and facilitating community engagement within OSS projects.

Chapter 4: *Characterizing Usability Issue Discussions in Open Source Software Projects*

After investigating various types of issues in three popular OSS projects, we discovered that there are specific formats and templates for posting technical issues. So, these findings led us to explore other types of issues discussed in ITSs. We discovered that UX and overall usability-related issues do not follow any specific formats, allowing authors to express their thoughts, beliefs and opinion more clearly without any predefined templates. As a result, we narrowed down the research to inquire about usability issues. This was the leading point for the next research, investigating usability dimensions in projects with Graphical User Interfaces (GUI). Therefore, in Chapter 4, we did a comprehensive study focusing on the often-neglected usability aspects in OSS projects. While OSS communities commonly use

ITSs (ITSs) for discussing usability concerns, little is known about the extent of these discussions, the specific usability dimensions addressed, and the collaborative dynamics involved. The research aims to fill this gap by examining usability discussions across five popular OSS projects. Findings reveal that while usability topics are prevalent, discussions primarily revolve around efficiency, flexibility, aesthetics and design. These issues are more frequently initiated by experienced community members and often involve visual communication and increased participant engagement. Moreover, the study identifies distinctive characteristics of usability discussions compared to non-usability topics. Understanding these dynamics can guide practical recommendations for OSS communities to enhance usability discussions and pave the way for future research on collaboration techniques within diverse communities, particularly concerning usability issues in the OSS context.

Chapter 5: *Analyzing Argument Discourse and Argument Quality in Open Source Usability Discussions*

Investigating, reading and annotating the issues and then annotating them as usability or non-related usability, as well as usability dimensions for distinguished usability issues, gave us a blurry image that there is a latent pattern in issue discussion threads. Since we investigated the sentiment and tone in the previous two studies, we went through the lens of argument mining and argument quality dimensions and investigated their impact on issue discussion threads. As a result, in Chapter 5, we investigated the significant role of argument discourse and quality in usability issue discussions. Argumentation plays a vital role in these discussions, enabling participants and collaborators to express their stances and opinions as well as persuade others in discussion threads. However, the dynamics and characteristics of argument discourse in these discussions have remained largely unexplored, hindering effective support for OSS usability discussions. To address this gap, a detailed analysis was conducted on argument discourse and quality in five OSS projects. The findings confirm that usability discussions predominantly revolve around arguments, with issue posts exhibiting higher-quality arguments compared to comments. These argumentative aspects significantly influence various discussion features, indicating their crucial role in OSS usability discussions. The research highlights the impact of argument discourse and quality, offering insights to bolster effective arguments and ultimately enhance OSS usability practices.

Chapter 6: *Characteristics of Usability Issues Posted by UX Professionals*

Regarding the three previous research studies, we hypothesized that the lack of designers' participation in the OSS community led to the neglecting of usability and low quality of